

HOMESTEAD MASTER PLAN

Optimized System for 2 People — Midlands, KwaZulu-Natal

Compressed edition: 11 sections, every fact retained

1. Sections & Loops

A section is a loop: one tank, one chemistry, all zones, mediums, animals, and humans it serves.

#	Loop	Chemistry	Zones	Mediums	Animals	Humans Get
1	Fish	Fish water (EC 0.8–1.2, pH 6.5–7.0)	Z1	Tanks, clarifier, biofilter, sump	Tilapia	—
2	Floating tray	Fish water	Z1	Floating trays	Duckweed, azolla (fed to tilapia)	—
3	Leaf	Leaf (EC 1.8, pH 6.1)	Z2, Z3, Z4	DWC raft + NFT gully + young Dutch buckets + isolated pot	Greens/herbs feed ducks, goats	Kale, chard, spinach, pak choi, coriander, basil, parsley, chives, spring onion, spearmint
4	Fruit	Fruit (EC 2.4, pH 5.9)	Z1, Z2	Fruiting Dutch buckets	Culls feed ducks, goats	Tomato, pepper, chilli, cucumber, zucchini
5	Tuber	Tuber (EC 1.5, pH 5.8)	Z1, Z4	Deep box + fabric bag	Vines feed ducks, goats	Sweet potato, potato
6	Moringa	Moringa (EC 1.2, pH 6.5)	Z1	40L pots	Prunings feed ducks, goats	Moringa leaf (dried, powdered)
7	Water	Water (EC 0–0.5, pH 6.0)	Z3	Nursery shelf	—	Seeds, cuttings
8	Duck	N/A	Z1	Deep litter; 30–40 m ²	Ducks	Duck meat, duck eggs
9	Goat	N/A	Z4	Dry bedding; 20–30 m ²	Goats	Goat meat, goat milk
10	BSF	N/A	Z1	Totes; 4–6 m ²	BSF larvae	—
11	Mulberry	N/A	Outdoor	Soil; 400–600 m ²	Mulberry	—

Space: GH ~230–330 m²; Outdoor ~1,300–1,850 m²; Total ~1,130–1,580 m² (duck/goat outdoor areas are the same mulberry hedges, counted once).

2. Chemistry

Line	EC	pH	N	P	K	Ca	Mg	N:K	Ca:Mg	Dosed?
Water	0–0.5	6.0	0	0	0	0	0	—	—	No
Leaf	1.8	6.1	170–200	40–50	200–230	140–160	40–50	0.85:1	3.2:1	Stock A+B + acid
Fruit	2.4	5.9	160–180	45–50	300–340	160–180	45–55	0.5:1	3.3:1	Stock A+B + acid + K
Tuber	1.5	5.8	95–115	42–48	210–240	115–120	37–40	0.45:1	3:1	Stock A+B + acid
Moringa	1.2	6.5	~100	~30	~120	~80	~25	0.85:1	3.2:1	Stock A+B + pH up
Fish water	0.8–1.2	6.5–7.0	30–100	10–30	little	—	—	—	—	Bicarbonate only

Rule: Change pH by no more than 0.3 per day. Keep calcium apart from sulfate and phosphate (Stock A and B). Use Fe-DTPA or Fe-EDDHA.

Control system: EC/pH probe per loop; bleed valve, dilution valve, dosing pump, pH pump, K boost valve (Fruit), emergency isolation, backup aeration, manual override.

3. Animals

Animal	Housing / Space	Eats	Produces	Feeds Into
Tilapia	3 tanks 1,200L; 60–80 m ²	Duckweed, azolla, BSF larvae	Fish, nitrogen	Fish loop > Leaf; humans
Ducks	Coop + run; 30–40 m ²	Mulberry, vines, moringa prunings, kale/chard/pak choi trimmings, culls, BSF larvae	Eggs, meat, manure	Humans, BSF, compost
Goats	Shelter; 20–30 m ²	Mulberry, vines, moringa prunings, browse	Milk, meat, manure	Humans, BSF, compost
BSF	4 totes + cage; 4–6 m ²	Kitchen waste, manure, fish solids	Larvae, frass	Fish, ducks; compost tea
Humans (2)	—	All of the above + tubers, greens, herbs, moringa	Kitchen waste, manure	BSF, compost

Humane slaughter: Tilapia — bolt pistol or percussive stun, then bleed. Ducks — non-penetrative captive bolt (convex head), then bleed. Goats — penetrative captive bolt, then bleed. Stun first, never bleed while conscious.

4. Feed Crops & Space

Species	Space	Zone	Medium	Loop	Feeds
Duckweed	40–60 m ²	Z1	Floating trays	Fish water	Tilapia
Azolla	10–15 m ²	Z1	Floating trays	Fish water	Tilapia
Mulberry	400–600 m ²	Outdoor	Soil, pollard hedges	N/A	Ducks, goats
Sweet potato vines (from tubers)	—	Z1+Z4	Coco:perlite 60:40	Tuber	Ducks, goats
Moringa prunings (from moringa)	—	Z1	Coco:perlite 60:40	Moringa	Ducks, goats

Mediums (8): Coco:perlite 60:40 (tubers, fruiting, moringa, spearmint); DWC raft + plug (greens); NFT gully + coco plug (herbs); floating trays (duckweed/azolla); water (tilapia); deep litter / dry bedding (ducks, goats); BSF totes; soil (mulberry).

5. Water Flow

- Source water > Fish loop (bicarbonate if pH < 6.5) > Fish sump > splits:
- **Fish sump > Floating trays > back to fish sump.** Bleed if EC > 1.2.
- **Fish sump > Leaf** (dose Stock A+B to EC 1.8, pH 6.1) > serves Z2/Z3/Z4 > return > bleed if EC > 2.0 or pH > 6.3 > dilute if needed > re-dose if EC < 1.5.
- **Leaf overflow > Fruit** (dilute, pH down, add K to EC 2.4, pH 5.9) > serves Z1/Z2 > return > bleed if EC > 2.6 or pH > 6.1 > dilute if needed > re-dose if EC < 2.0.
- **Fruit overflow > Tuber** (dilute, pH down to EC 1.5, pH 5.8) > serves Z1/Z4 > return > bleed if EC > 1.7 or pH > 6.0 > dilute if needed > re-dose if EC < 1.3.
- **Tuber overflow > Moringa** (dilute, pH up to EC 1.2, pH 6.5) > serves Z1 > drain to waste > compost tea / garden.
- **Water loop > nursery in Z3** > return > no dose > spent to garden.
- All spent water > garden / soil / compost tea. Soil never returns.

Sequence: Dose at start, bleed when out of range, dilute if needed, re-dose only if EC drops.

Hard rules: Dosed water never returns to fish. Different loops never mix. Moringa never shares. Soil and drain-to-waste never return. Spent water goes to garden only. Change pH max 0.3/day.

6. Layout

Z1 Tropical	Z2 Warm	Z3 Herb Bench	Z4 Cool
Floating trays	Fruit Dutch buckets	Nursery shelf	DWC raft + NFT gully
Sweet potato deep box	Tomato, pepper	Basil, parsley	Kale, chard, spinach
Moringa 40L pots	Chilli	Chives, spring onion	Pak choi, coriander
Fruit Dutch buckets	Cucumber	Spearmint pot	Potato fabric bags
Duck coop	Zucchini	—	Goat shelter
BSF totes	—	—	—

Utility strip: Fish tanks > Clarifier > Biofilter > Sump > Recipe tanks.

7. Plants: Stages & Staggered Sowing

Sow Every	Species	#	Stage 1	Stage 2	Stage 3	Stage 4
1 wk	Pak choi	3	Water Z3	Leaf Z4	Leaf Z4	—
1 wk	Spring onion	2	Water Z3	Leaf Z3	Leaf Z3	—
2 wk	Spinach	6	Water Z3	Leaf Z4	Leaf Z4	Leaf Z4
2 wk	Coriander	2	Water Z3	Leaf Z4	Leaf Z4	—
4 wk	Kale	3	Water Z3	Leaf Z4	Leaf Z4	Leaf Z4
4 wk	Swiss chard	3	Water Z3	Leaf Z4	Leaf Z4	Leaf Z4
4 wk	Sweet basil	2	Water Z3	Leaf Z3	Leaf Z3	—
4 wk	Sweet potato	5	Water Z3	Tuber Z1	Tuber Z1	Tuber Z1 (flush)
4 wk	White potato	3	Water Z3	Tuber Z4	Tuber Z4	Tuber Z4 (flush)
5 wk	Zucchini	2	Water Z3	Leaf Z2	Leaf Z2	Fruit Z2
8 wk	Tomato	5	Water Z3	Leaf Z2	Leaf Z2	Fruit Z2
8 wk	Cucumber	2	Water Z3	Leaf Z2	Leaf Z2	Fruit Z2
12 wk	Bell pepper	6	Water Z3	Leaf Z2	Leaf Z2	Fruit Z2
13 wk	Parsley	3	Water Z3	Leaf Z3	Leaf Z3	Leaf Z3
13 wk	Moringa	2	Water Z3	Leaf Z1	Moringa Z1	Moringa Z1
16 wk	Chilli	2	Water Z3	Leaf Z2/Z1	Leaf Z2/Z1	Fruit Z2/Z1
26 wk	Garlic chives	2	Water Z3	Leaf Z3	Leaf Z3	Leaf Z3
26 wk	Spearmint	2 pots	Water Z3	Leaf Z3	Leaf Z3	Leaf Z3
Continuous	Duckweed	half/3 days	Fish loop Z1	Fish loop Z1	Fish loop Z1	Fish loop Z1
Continuous	Azolla	half/3 days	Fish loop Z1	Fish loop Z1	Fish loop Z1	Fish loop Z1

Seed quantities per sowing: Pak choi 5–6; spring onion 4–5; spinach 10–12; coriander 4–5; kale 5–6; chard 5–6; basil 4–5; tomato 8–10; cucumber 4–5; pepper 10–12; chilli 4–5; zucchini 4–5; parsley 5–6; moringa 4–5; chives 4–5; spearmint cuttings.

Animal rhythm: Every 1 wk — new BSF tote, collect eggs daily. Every 2 wk — harvest half duckweed/azolla. Every 6 wk — breed next goat doe. Every 8 wk — add 35 tilapia, harvest from tank 3. Every 26 wk — replace 2 ducks. Continuous — ducks breed, goats kid yearly.

Week rhythm: Mon sow/plug; Tue move batches to next loop; Wed EC/pH check every tank; Thu harvest; Sat cut animal greens and prune moringa.

8. Diet (2 People, Per Day)

Source	Amount	Protein	Fat	Key Nutrients
Tilapia	250–400g	50–80g	5–8g	Omega-3, vitamin D
Duck meat	125–275g	25–55g	12–28g	B12, iron, DHA
Duck eggs	3–6 eggs	18–36g	15–30g	B12, choline, iron
Goat meat	100–200g	20–40g	3–6g	B12, zinc, iron
Goat milk	1–2L	30–60g	30–60g	Calcium, zinc, B12
Moringa	50g dried	13g	3g	Calcium, zinc, iron
Tubers	1.5kg	20g	2g	Carbs, potassium
Greens	400g	8g	2g	Vitamins A, C, K
TOTAL	—	184–312g	70–139g	Complete

Targets met: 2,500 kcal, 150g protein, ~312g carbs, 10%+ fat. Moringa covers calcium and zinc. Animal foods cover B12, iron, omega-3.

9. Cost & Solar

Item	Low	High
Capital (DIY, solar, greenhouse)	R333,000	R570,000

Item	Low	High
Annual running	R9,500	R37,500

Solar load: Winter 12–21 kWh/day; Summer 2–4 kWh/day. Recommended: 8–10 kWp panels + 20 kWh battery + 5kW inverter + CoC. Generator backup 3kW.

Buy list: Salt + iodine (cannot grow); cooking oil (unless oilseed crops added); vitamin D3 winter (sunlight insufficient); spices (black pepper, cumin).

10. Storage

Item	Method	Shelf Life
Potatoes	Cure 15–20°C, store 4–7°C dark	4–5 months
Sweet potatoes	Cure 30–32°C, store 12–14°C	4–6 months
Dried moringa	Air-dry, powder, airtight	6–12 months
Dried herbs/chilli	Air-dry, airtight	6–12 months
Frozen meat/fish	Freeze	6–12 months
Canned meat/fish	Pressure can	1–5 years
Eggs	Fridge	4–5 weeks
Duckweed/azolla	Use fresh	1–2 days

11. Summary

Structure: 11 sections = 7 loops + 4 animal/fodder sections. 6 chemistry lines, 8 mediums, 4 zones + outdoor, 15+ plants, 4 animals, 2 humans fed. Total ~1,130–1,580 m² (GH ~230–330 m²).

Water: Fish > Floating trays > Fish. Fish > Leaf > Fruit > Tuber > Moringa > Soil. Dose at start, bleed when out of range, dilute if needed, re-dose only if EC drops. Nothing dosed returns to fish. Soil never returns.

Animals: Tilapia in fish loop. Ducks and goats eat trimmings, produce eggs/milk/meat/manure. BSF converts waste to feed. Humans eat everything; waste feeds BSF.

Output: Feeds 2 people 2,500 kcal and 150g protein daily. All animal feed grown on site. Only salt, iodine, oil, and winter vitamin D bought.